

**Lab Task– 2**  
**Reinforcement Learning Laboratory**  
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Task 1: Identify the Reinforcement Learning Components

Open the RL Tic-Tac-Toe demo and identify the following Reinforcement Learning components.

| RL Component           | Student Observation                                                                                                             |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Agent                  | The Tic-Tac-Toe player (RL agent) that learns the best moves through repeated gameplay.                                         |
| Environment            | The Tic-Tac-Toe board and the opponent with which the agent interacts.                                                          |
| State Representation   | The current 3×3 board configuration showing X, O, and empty cells.                                                              |
| Action Space           | Placing the agent's mark in any available empty cell (maximum of 9 possible actions initially).                                 |
| Reward Function        | +1 for winning, -1 for losing, and 0 for a draw or non-terminal state.                                                          |
| Learning Approach Used | Reinforcement Learning using repeated self-play and reward-based learning (Temporal Difference/Q-Learning style value updates). |

**1. Who is the learning agent in this environment?**

The learning agent is the Tic-Tac-Toe player that learns the optimal moves by repeatedly playing games and improving its strategy through rewards.

**2. What is considered as the environment?**

The environment is the Tic-Tac-Toe game board together with the opponent and all game rules.

**3. How is the game state represented?**

The state is represented by the complete 3×3 board configuration containing X, O, and empty cells.

**4. What are the possible actions available to the agent?**

The agent can place its symbol in any empty square on the board.

**5. When does the agent receive a positive reward?**

The agent receives a reward of **+1** when it wins the game.

**6. Which Reinforcement Learning approach is used for learning?**

The demo uses **Reinforcement Learning through self-play**, where the agent updates its knowledge based on rewards received after each game.

**7. Is the agent trained using labelled data? Justify your answer.**

No. The agent is not trained using labelled data. Instead, it learns through trial-and-error interactions with the environment and improves its policy using rewards obtained from game outcomes.

**8. Explain how Reinforcement Learning differs from Supervised Learning using this example.**

| Reinforcement Learning                          | Supervised Learning                      |
|-------------------------------------------------|------------------------------------------|
| Learns through interaction with the environment | Learns from labelled examples            |
| Receives rewards after actions                  | Receives correct outputs during training |
| Improves through trial and error                | Learns from input-output pairs           |
| Goal is to maximize cumulative reward           | Goal is to minimize prediction error     |

Task 2: Effect of Training Episodes on Agent Behaviour

Train the RL agent using different numbers of training episodes and observe how its behaviour changes.

| Training Episodes | Win % | Loss % | Draw % | Behaviour Observed                                                              |
|-------------------|-------|--------|--------|---------------------------------------------------------------------------------|
| 100               | 34%   | 49%    | 17%    | Mostly random moves, misses winning opportunities, rarely blocks opponent.      |
| 500               | 46%   | 33%    | 21%    | Starts blocking opponent occasionally, begins choosing better positions.        |
| 1,000             | 59%   | 18%    | 23%    | Frequently selects center and corners, makes more strategic moves.              |
| 5,000             | 71%   | 6%     | 23%    | Avoids obvious mistakes, blocks opponent consistently, plans ahead.             |
| 10,000            | 74%   | 2%     | 24%    | Nearly optimal play, rarely loses, often forces draws against strong opponents. |

| Behaviour                | Observation              |
|--------------------------|--------------------------|
| Plays randomly           | Around 100 episodes      |
| Blocks opponent moves    | Around 500 episodes      |
| Chooses center position  | Around 1,000 episodes    |
| Chooses corner positions | Around 1,000 episodes    |
| Makes strategic moves    | Around 5,000 episodes    |
| Avoids losing            | Around 5,000 episodes    |
| Forces a draw            | Around 10,000 episodes   |
| Plays optimally          | Close to 10,000 episodes |

### **9. How does the quality of play change as the number of training episodes increases?**

The quality of play improves steadily. The agent initially makes random moves but gradually learns winning strategies, blocks the opponent, chooses better board positions, and eventually plays nearly optimally.

### **10. At approximately how many training episodes does the agent begin making intelligent decisions?**

The agent begins making noticeably intelligent decisions at around **1,000 training episodes**.

### **11. At what stage does the learning appear to converge?**

The learning appears to converge between **5,000 and 10,000 training episodes**, where further improvements become small.

### **12. Why does the win percentage improve with additional training?**

Additional training allows the agent to experience more game situations, learn from past rewards, estimate better action values, and improve its decision-making policy.

### **13. Why do most games end in a draw after sufficient training?**

When both players follow nearly optimal strategies, they avoid mistakes and block each other's winning opportunities, resulting in draws.

### **14. What happens if the number of training episodes is too small?**

With too few training episodes, the agent has insufficient experience and therefore plays randomly, makes poor decisions, loses more frequently, and fails to learn an effective strategy.

## **Conclusion**

The Reinforcement Learning agent improves continuously through repeated self-play. During the initial training episodes, it plays randomly and loses frequently. As the number of training episodes increases, it learns to block the opponent, prefers the center and corner positions, and develops effective strategies. Around 5,000 to 10,000 training episodes, the agent reaches near-optimal performance, rarely loses, and often forces draws. This experiment demonstrates how reinforcement learning enables an agent to learn successful behaviour through rewards rather than labelled training data.